AAP-003 — Height, Breadth, and Depth
A Three-Dimensional Framework for AI Evolution

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Repository: https://github.com/sizhet/AI-Action-Paths-AAP

Abstract

Artificial intelligence has traditionally been evaluated by a single dominant objective: increasing capability.
Model scaling, benchmark performance, reasoning ability, coding accuracy, and scientific problem solving have all contributed to one primary direction—making AI increasingly intelligent.
AAP argues that this one-dimensional perspective is no longer sufficient.
As AI begins to participate in science, engineering, education, industry, healthcare, robotics, and society, future progress depends not only on **how intelligent AI becomes**, but also on **how many meaningful actions it can perform** and **how deeply it can continuously participate**.
This chapter proposes a three-dimensional framework for understanding AI evolution:

- **Height** — Intelligence Capability
- **Breadth** — Action Diversity
- **Depth** — Sustainable Participation

Together they form the HBD Framework.

From One-Dimensional Growth to Three-Dimensional Evolution

The first generation of AI development was naturally driven by capability.
Higher capability enabled AI to solve increasingly difficult problems.
This strategy produced remarkable achievements, particularly in Large Language Models.
However, practical deployment has revealed another reality.
High intelligence alone does not automatically produce broad societal impact.
Real-world AI systems must also:

- participate in many different activities,

- operate continuously,
- accumulate experience,
- earn trust,
- collaborate with humans,
- integrate into organizations.

These requirements extend beyond intelligence alone.
AAP therefore views AI evolution as a balanced optimization problem across three complementary dimensions.

Height
Intelligence Capability

Height measures how capable an AI system becomes.
Typical examples include:

- reasoning ability,
- knowledge acquisition,
- coding capability,
- planning,
- mathematical performance,
- scientific problem solving,
- benchmark scores.

Height answers one question:
How intelligent can AI become?
Large Language Models have dramatically expanded this dimension.
Model scaling remains one of the most successful strategies for increasing Height.
AAP fully recognizes Height as an essential component of future AI evolution.

Breadth
Action Diversity

Breadth measures the diversity of meaningful actions AI can reliably perform.
Examples include:

- discussion,
- software engineering,
- documentation,
- education,
- robotics,
- healthcare,
- research assistance,
- governance,
- autonomous services.

Breadth answers a different question:
In how many ways can AI participate?
Unlike Height, which measures capability, Breadth measures participation.
As AI enters more domains, Action Space continuously expands.
AAP considers this expansion to be one of the defining characteristics of future AI systems.

Depth
Sustainable Participation

Depth measures how continuously, autonomously, and reliably AI can perform an action over time.
Examples include:

- persistent runtime,
- memory,
- trusted execution,
- Runtime Invariants,
- long-term task execution,
- self-monitoring,
- continuous improvement,
- organizational integration.

Depth answers another question:
How deeply can AI work?
Many current AI systems perform isolated tasks extremely well.
Future AI systems will increasingly perform long-running responsibilities.
Brain Units represent one important direction toward increasing Depth.

The HBD Space

The three dimensions together define a conceptual HBD Space.
Every AI system can be positioned somewhere within this three-dimensional space.
Different Action Paths emphasize different dimensions.
For example,

- LLM Scaling primarily expands Height.
- Human–AI Discussion expands Breadth.
- Brain Units significantly increase Depth.
- Known Gap Workers emphasize Breadth with moderate Depth.
- Unknown Gap Workers combine high Height with increasing Breadth and Depth.
- Hybrid Organizations seek balanced optimization across all three dimensions.

Rather than competing, these Action Paths complement one another by expanding different regions of HBD Space.

Why Balance Matters

Optimizing only one dimension eventually encounters diminishing returns.
For example,
High Height without Breadth creates extremely intelligent systems with limited participation.
High Breadth without sufficient Depth produces many disconnected capabilities that cannot sustain long-term value.
High Depth without adequate Height limits problem-solving capability.
Long-term AI evolution therefore requires coordinated growth across all three dimensions.
AAP refers to this objective as **Balanced HBD Optimization**.

Relation to AI Action Paths

The HBD Framework provides a geometric interpretation of AI Action Paths.
Each Action Path expands AI evolution in a different direction.
LLM Scaling increases Height.
Human–AI Discussion expands Breadth.
Brain Units strengthen Depth.
Known Gap Workers increase practical participation.
Unknown Gap Workers push the frontier of scientific discovery.
Hybrid Organizations integrate all three dimensions simultaneously.
Consequently, Action Paths should not be viewed as isolated technologies.
They represent coordinated strategies for expanding HBD Space.

Beyond Intelligence

One important implication follows naturally.
Future AI development is no longer a race toward a single numerical objective.
Instead, it becomes a balanced optimization problem involving intelligence, participation, and sustained execution.
This shift changes the central question from
How intelligent can AI become?
to
How intelligently, how broadly, and how deeply can AI participate in the real world?
This perspective complements existing benchmark-driven evaluation while introducing a new organizational understanding of AI evolution.

Key Takeaways

- AI evolution is no longer adequately described by intelligence alone.
- Height measures capability and reasoning performance.
- Breadth measures the diversity of meaningful AI actions.
- Depth measures sustainable and trusted long-term participation.
- Different AI Action Paths expand different regions of HBD Space.
- Future AI systems should pursue balanced optimization across Height, Breadth, and Depth.
- The HBD Framework complements traditional benchmark-oriented evaluation with an action-oriented perspective.
- Balanced HBD Optimization provides a roadmap for scalable Human–AI participation.